

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: titanic = sns.load_dataset('titanic')
```

```
In [5]: titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   class       891 non-null    category
9   who         891 non-null    object
10  adult_male   891 non-null    bool
11  deck        203 non-null    category
12  embark_town  889 non-null    object
13  alive        891 non-null    object
14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

```
In [6]: titanic.head()
```

```
Out[6]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
--	----------	--------	-----	-----	-------	-------	------	----------	-------	-----	---------

0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T



```
In [7]: titanic.head(10)
```

Out[7]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	
5	0	3	male	NaN	0	0	8.4583	Q	Third	man	
6	0	1	male	54.0	0	0	51.8625	S	First	man	
7	0	3	male	2.0	3	1	21.0750	S	Third	child	
8	1	3	female	27.0	0	2	11.1333	S	Third	woman	
9	1	2	female	14.0	1	0	30.0708	C	Second	child	

In [10]: `titanic.tail()`

Out[10]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult
886	0	2	male	27.0	0	0	13.00	S	Second	man	
887	1	1	female	19.0	0	0	30.00	S	First	woman	
888	0	3	female	NaN	1	2	23.45	S	Third	woman	
889	1	1	male	26.0	0	0	30.00	C	First	man	
890	0	3	male	32.0	0	0	7.75	Q	Third	man	

In [11]: `titanic.describe()`

Out[11]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [13]: titanic.describe().transpose()
```

```
Out[13]:
```

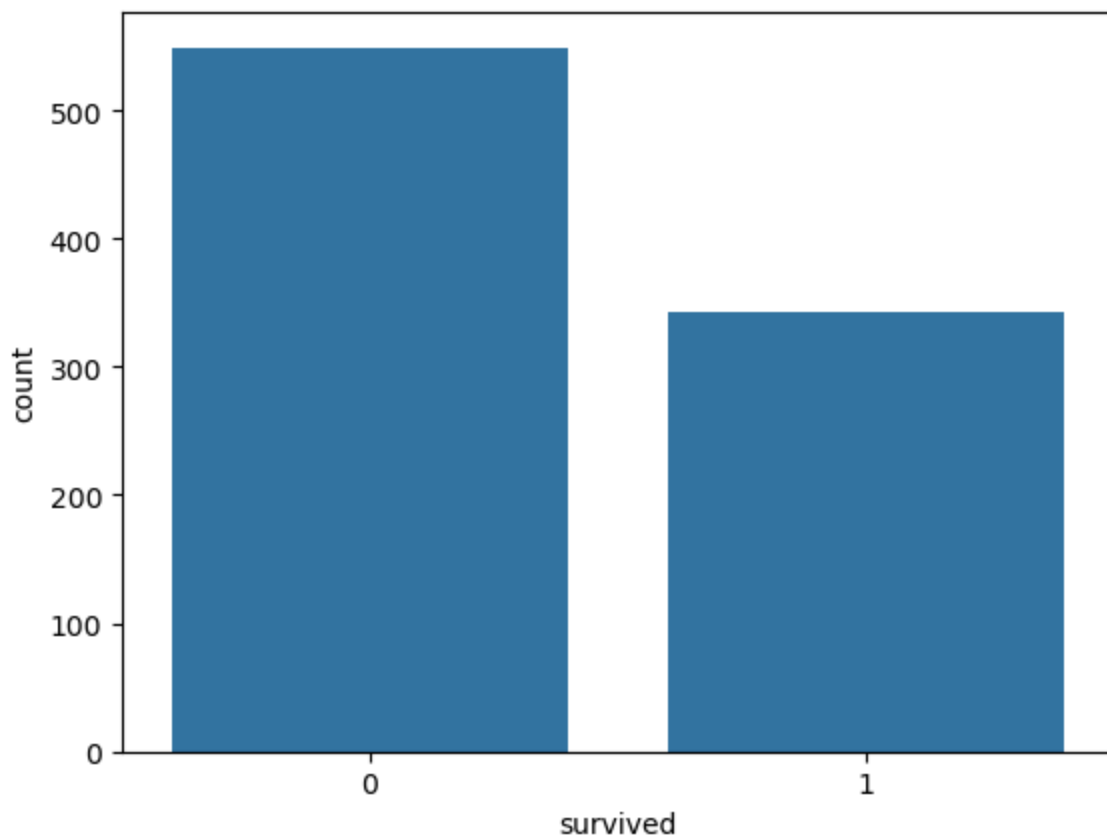
	count	mean	std	min	25%	50%	75%	max
survived	891.0	0.383838	0.486592	0.00	0.0000	0.0000	1.0	1.0000
pclass	891.0	2.308642	0.836071	1.00	2.0000	3.0000	3.0	3.0000
age	714.0	29.699118	14.526497	0.42	20.1250	28.0000	38.0	80.0000
sibsp	891.0	0.523008	1.102743	0.00	0.0000	0.0000	1.0	8.0000
parch	891.0	0.381594	0.806057	0.00	0.0000	0.0000	0.0	6.0000
fare	891.0	32.204208	49.693429	0.00	7.9104	14.4542	31.0	512.3292

```
In [14]: titanic['survived'].value_counts()
```

```
Out[14]: survived
0      549
1      342
Name: count, dtype: int64
```

```
In [16]: sns.countplot(data=titanic, x='survived')
```

```
Out[16]: <Axes: xlabel='survived', ylabel='count'>
```



```
In [35]: titanic['pclass'].value_counts()
```

```
Out[35]: pclass
3      491
1      216
2      184
Name: count, dtype: int64
```

```
In [36]: p_class = titanic[['pclass', 'class']]
p_class.head()
```

```
Out[36]:
```

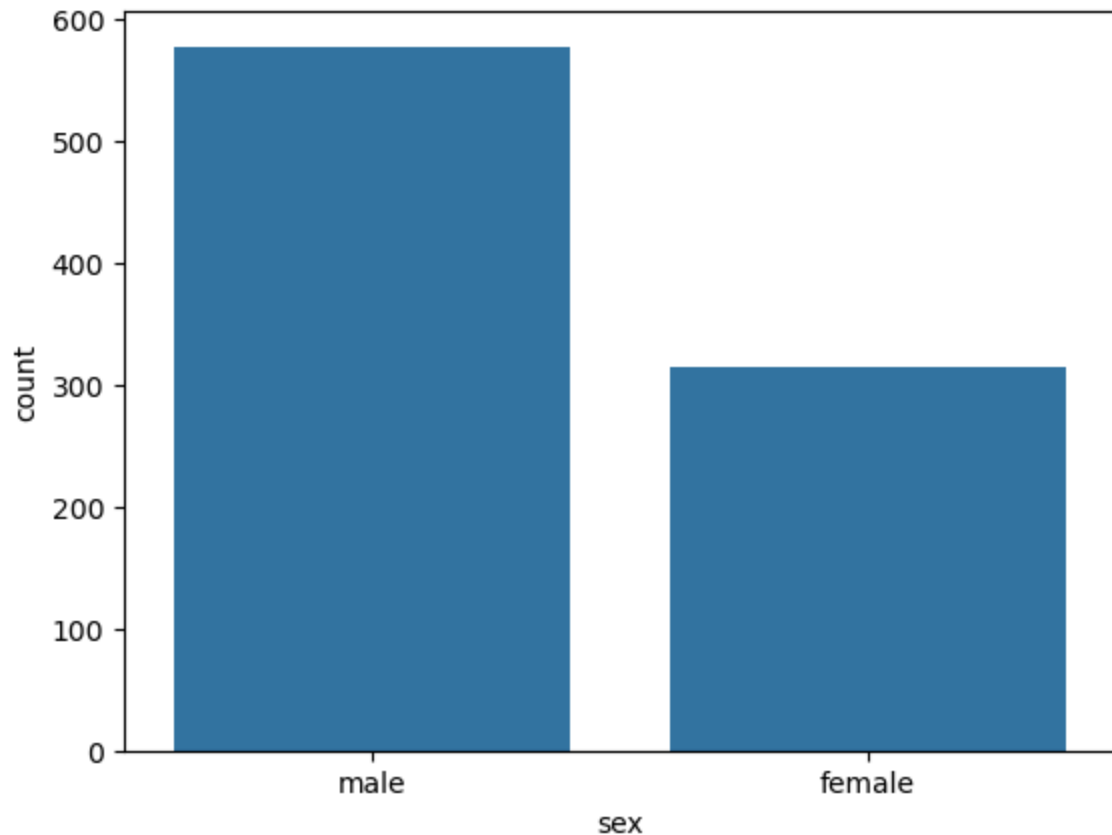
	pclass	class
0	3	Third
1	1	First
2	3	Third
3	1	First
4	3	Third

```
In [37]: titanic['sex'].value_counts()
```

```
Out[37]: sex
male      577
female    314
Name: count, dtype: int64
```

```
In [38]: sns.countplot(data=titanic, x='sex')
```

```
Out[38]: <Axes: xlabel='sex', ylabel='count'>
```



```
In [50]: less_than_20 = titanic[titanic['age'] < 20]
less_than_20.head()
```

```
Out[50]:
```

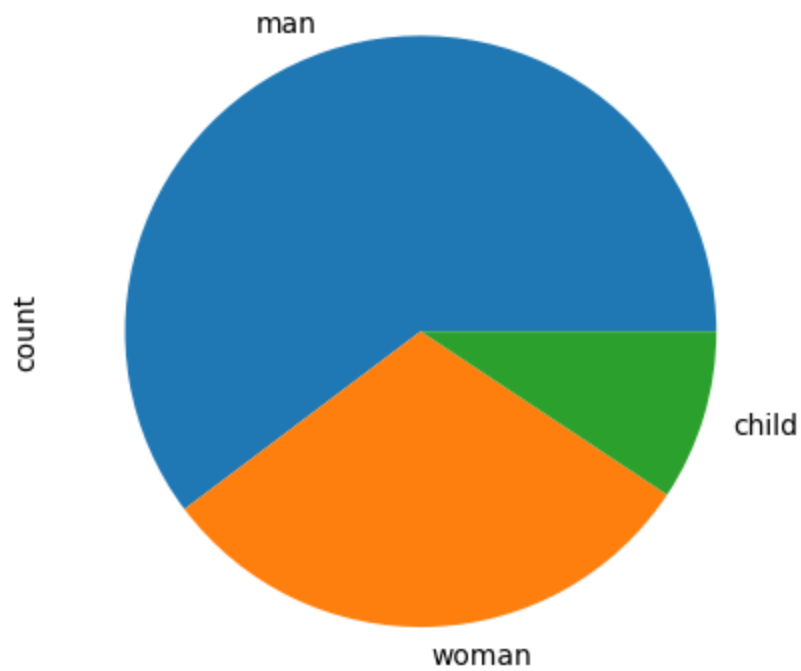
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_r
7	0	3	male	2.0	3	1	21.0750	S	Third	child	f
9	1	2	female	14.0	1	0	30.0708	C	Second	child	f
10	1	3	female	4.0	1	1	16.7000	S	Third	child	f
14	0	3	female	14.0	0	0	7.8542	S	Third	child	f
16	0	3	male	2.0	4	1	29.1250	Q	Third	child	f

```
In [51]: len(less_than_20)
```

```
Out[51]: 164
```

```
In [4]: titanic.who.value_counts().plot(kind='pie')
```

```
Out[4]: <Axes: ylabel='count'>
```

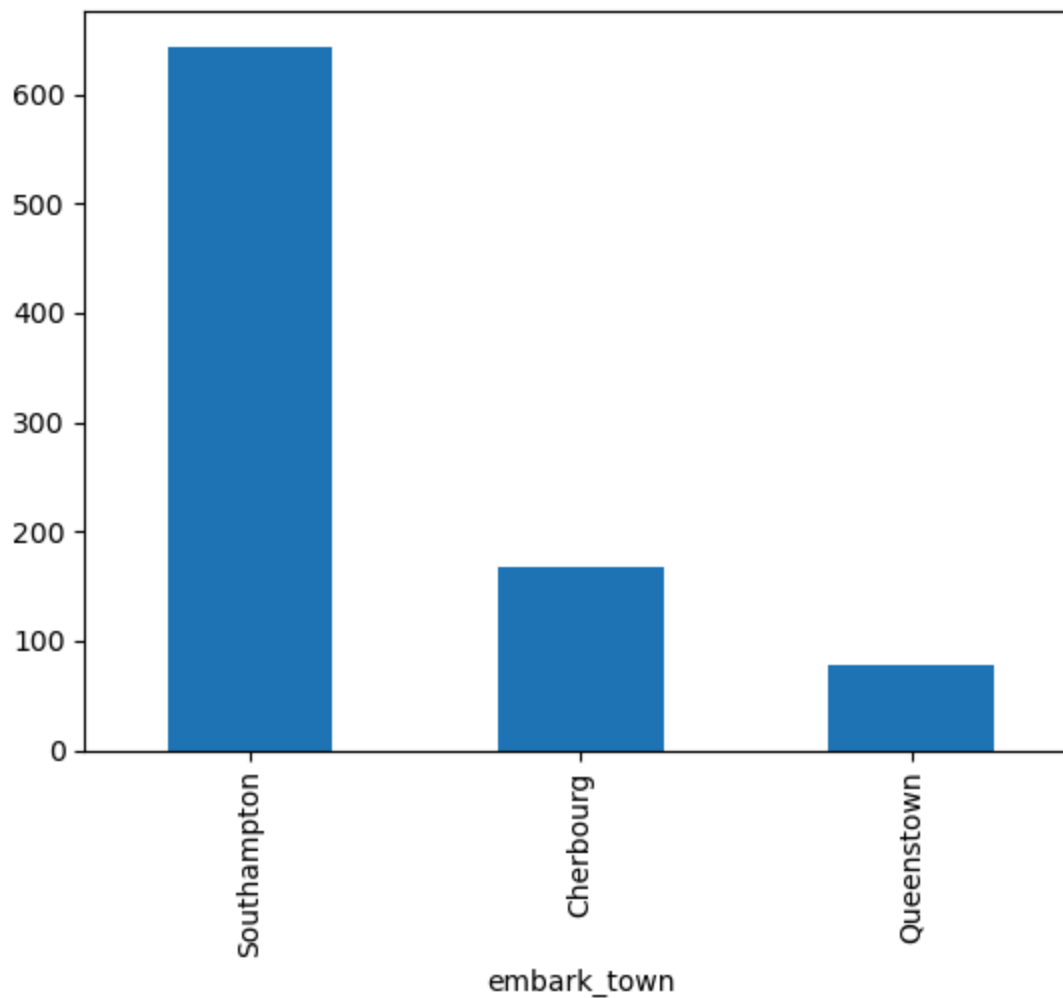


```
In [17]: titanic['embark_town'].nunique()
```

```
Out[17]: 3
```

```
In [12]: titanic['embark_town'].value_counts().plot(kind='bar')
```

```
Out[12]: <Axes: xlabel='embark_town'>
```

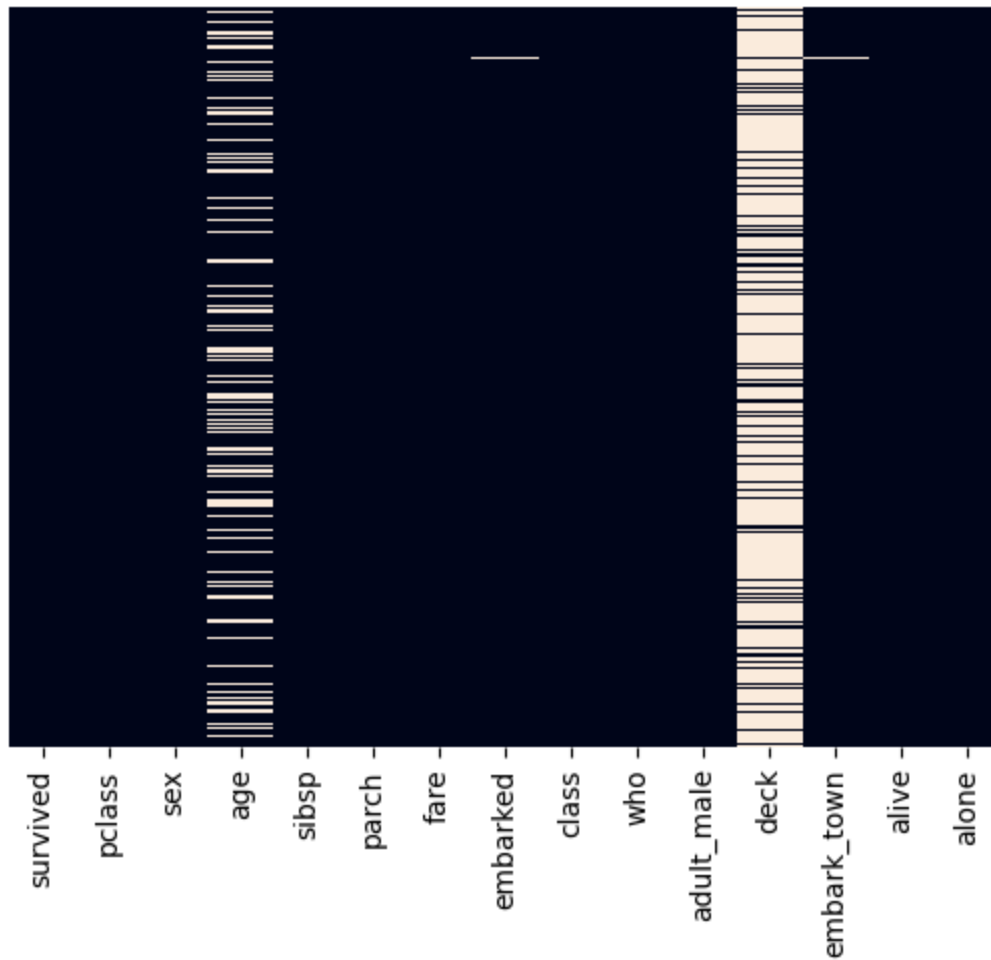


```
In [18]: titanic.isnull().sum()
```

```
Out[18]: survived      0
pclass      0
sex         0
age        177
sibsp      0
parch      0
fare       0
embarked    2
class      0
who        0
adult_male  0
deck       688
embark_town 2
alive      0
alone      0
dtype: int64
```

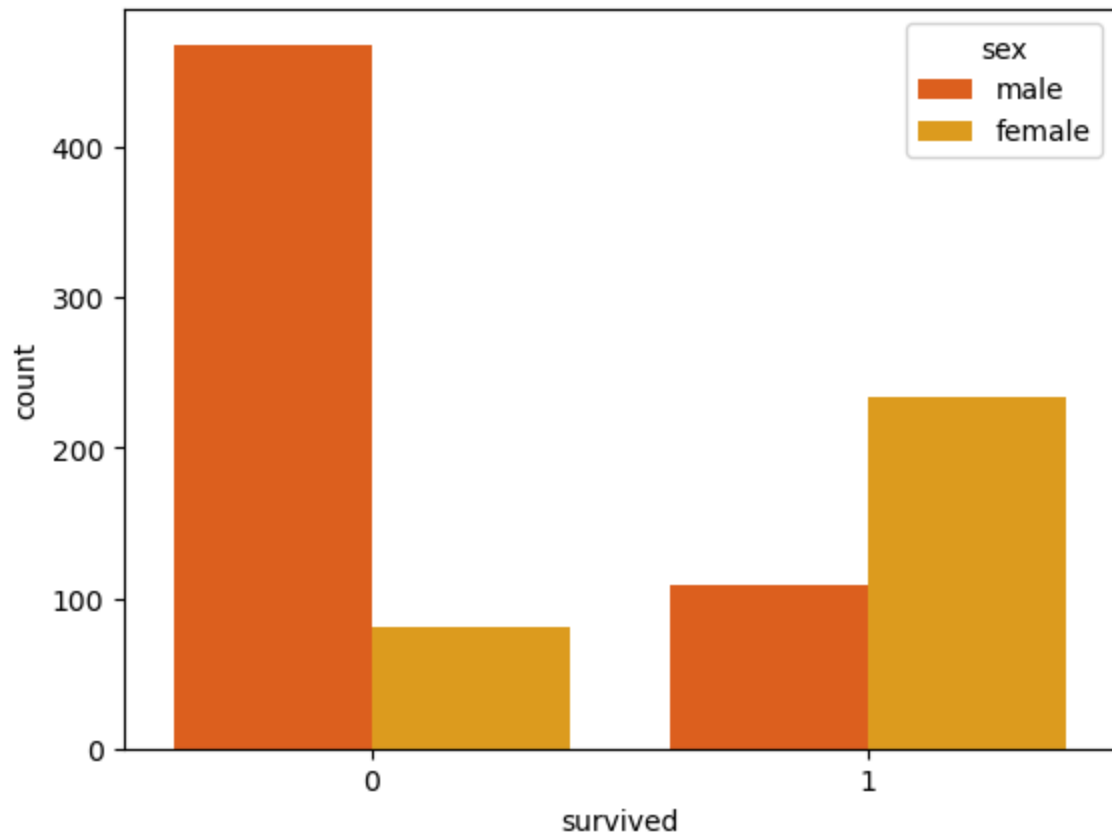
```
In [19]: sns.heatmap(titanic.isnull(),yticklabels=False,cbar=False)
```

```
Out[19]: <Axes: >
```



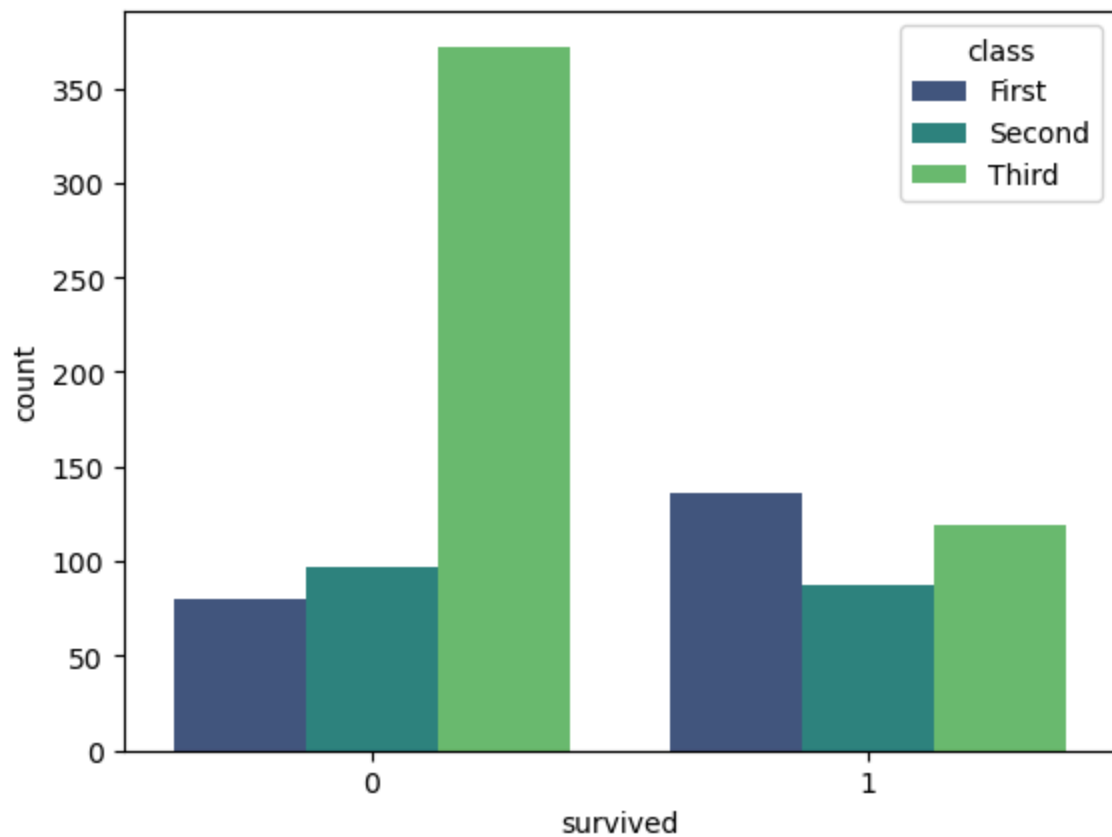
```
In [25]: sns.countplot(data=titanic, x='survived', palette='autumn', hue='sex')
```

```
Out[25]: <Axes: xlabel='survived', ylabel='count'>
```



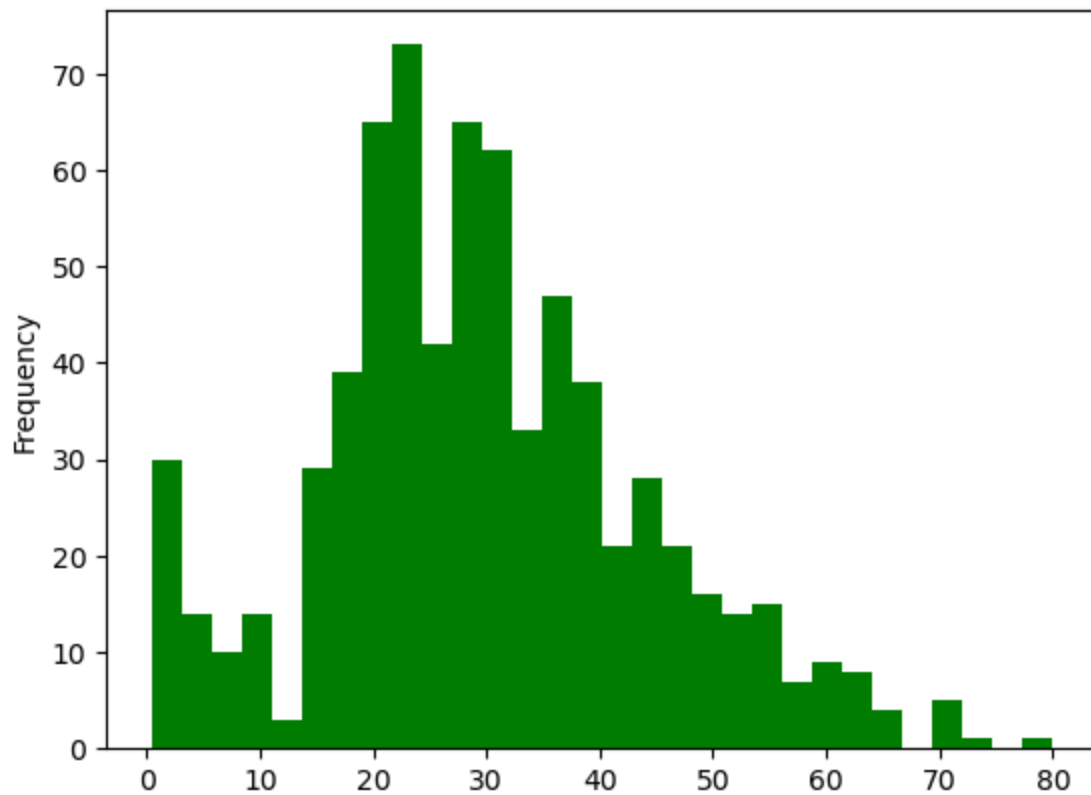
```
In [27]: sns.countplot(data=titanic, x='survived', palette='viridis', hue='class')
```

```
Out[27]: <Axes: xlabel='survived', ylabel='count'>
```



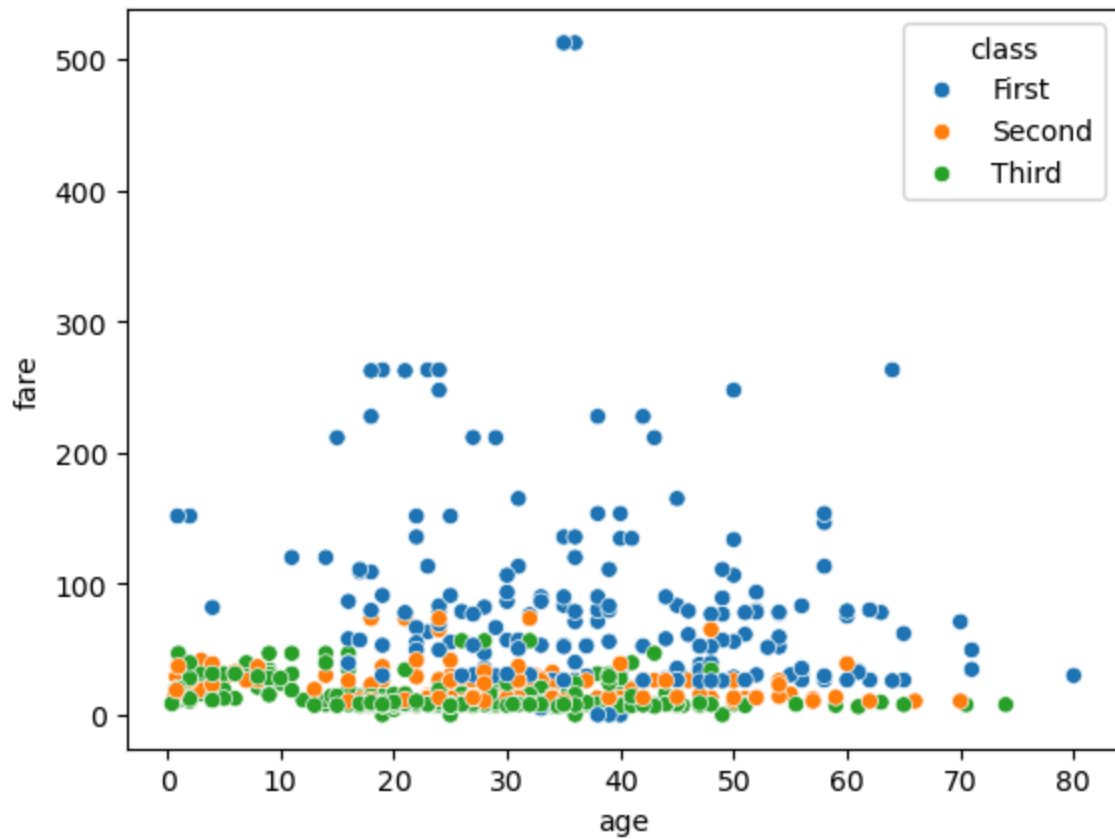
```
In [28]: titanic.age.plot(kind='hist',bins=30,color='green')
```

```
Out[28]: <Axes: ylabel='Frequency'>
```



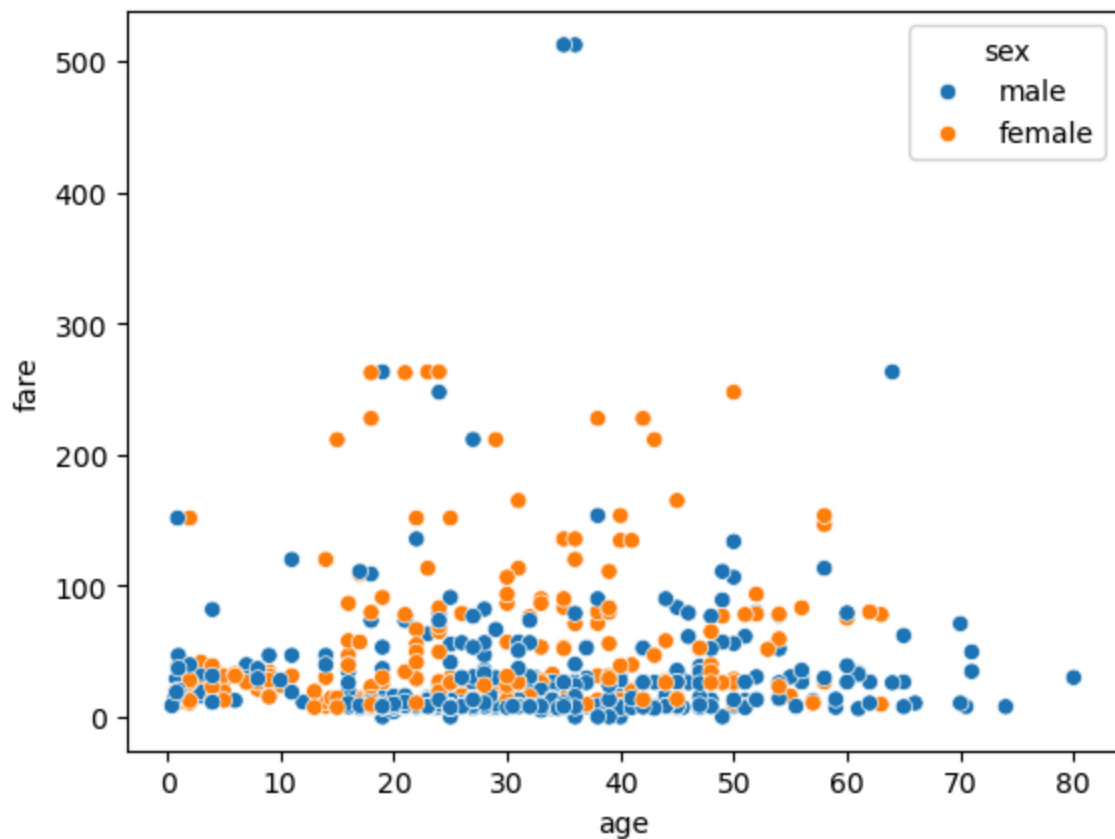
```
In [29]: sns.scatterplot(data=titanic, x='age', y='fare', hue='class')
```

```
Out[29]: <Axes: xlabel='age', ylabel='fare'>
```



```
In [30]: sns.scatterplot(data=titanic, x='age', y='fare', hue='sex')
```

```
Out[30]: <Axes: xlabel='age', ylabel='fare'>
```



```
In [35]: correlation = titanic.corr(numeric_only=True)
correlation
```

```
Out[35]:
```

	survived	pclass	age	sibsp	parch	fare	adult_male	
survived	1.000000	-0.338481	-0.077221	-0.035322	0.081629	0.257307	-0.557080	-0
pclass	-0.338481	1.000000	-0.369226	0.083081	0.018443	-0.549500	0.094035	0
age	-0.077221	-0.369226	1.000000	-0.308247	-0.189119	0.096067	0.280328	0
sibsp	-0.035322	0.083081	-0.308247	1.000000	0.414838	0.159651	-0.253586	-0
parch	0.081629	0.018443	-0.189119	0.414838	1.000000	0.216225	-0.349943	-0
fare	0.257307	-0.549500	0.096067	0.159651	0.216225	1.000000	-0.182024	-0
adult_male	-0.557080	0.094035	0.280328	-0.253586	-0.349943	-0.182024	1.000000	0
alone	-0.203367	0.135207	0.198270	-0.584471	-0.583398	-0.271832	0.404744	1

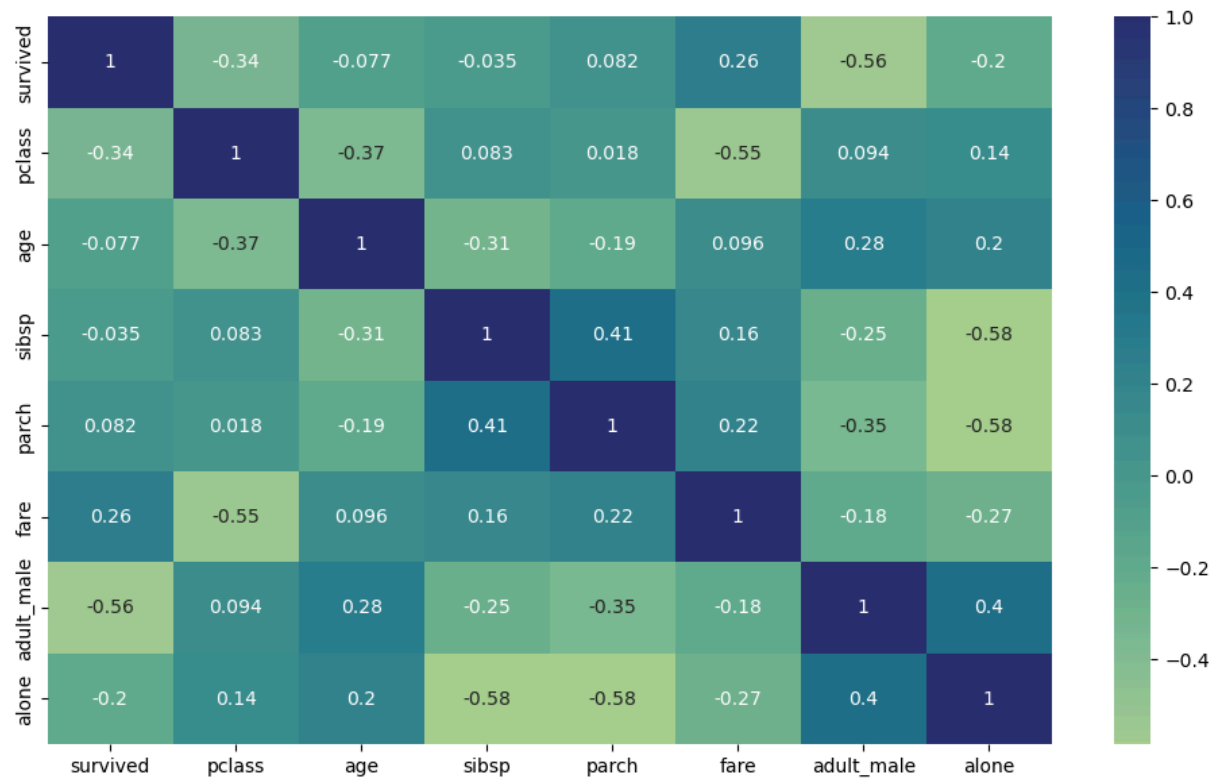


```
In [36]: correlation['survived']
```

```
Out[36]: survived      1.000000
pclass      -0.338481
age         -0.077221
sibsp       -0.035322
parch        0.081629
fare         0.257307
adult_male  -0.557080
alone       -0.203367
Name: survived, dtype: float64
```

```
In [37]: plt.figure(figsize=(12,7))
sns.heatmap(correlation,annot=True,cmap='crest')
```

```
Out[37]: <Axes: >
```



```
In [ ]:
```